Vivekanand Education Society's Institute of Technology



Department of Computer Engineering

Group No.: 17
Date :- 13-02-2026

Capstone Project - I Synopsis (2025-26) - Sem VI

Smart Monitoring of Jasmine Plant Growth, Disease Detection, and Yield Improvement

Dr. Rohini Temkar
Assistant Professor, CMPN

Sakshi Thorat
V.E.S.I.T
2023.sakshi.thorat@ves.ac.in

Pradnya Patil
V.E.S.I.T
2023.pradnya.patil@ves.ac.i

Diksha Patkar
V.E.S.I.T
2023.diksha.patkar@ves.ac.in

Akritee Singh
V.E.S.I.T
d2023.akritee.singh@ves.ac.in

Abstract

Jasmine cultivation in India plays a vital role in supporting rural livelihoods and contributing to the floriculture sector. However, farmers face several challenges that affect crop productivity and quality. These include late and inaccurate detection of plant diseases, dependence on subjective manual inspection, and the absence of integrated technological support for farm management. As a result, diseases often spread unnoticed, improper treatments are applied, and optimal cultivation practices are not followed, leading to significant losses in yield and income.

To address these limitations, this report proposes an AI and Machine Learning–based smart monitoring system specifically designed for jasmine plants. The system utilizes image-based disease detection techniques to identify infections at different growth stages, including leaves, buds, and flowers. It also estimates disease severity and provides AI-powered recommendations for timely treatment. In addition, the system incorporates yield prediction models to help farmers anticipate production levels and plan resources efficiently. By combining data analytics with advanced image processing, the proposed solution enables early diagnosis and precise decision-making.

Furthermore, the smart monitoring system integrates features such as market access information, fertiliser optimisation, and intelligent irrigation management. These components help farmers adopt sustainable agricultural practices while reducing input costs and environmental impact. By offering timely, accurate, and localized insights, the system empowers farmers, field supervisors, and agri-tech organizations to make informed, data-driven decisions. Ultimately, this approach enhances crop health, improves productivity, minimizes economic losses, and promotes the long-term sustainability of jasmine cultivation in India.

1. Introduction

Agriculture forms the backbone of India's rural economy, supporting a large segment of the population and ensuring national food security. Despite its importance, farmers continue to face numerous challenges, including unpredictable climatic conditions, water scarcity, fragmented landholdings, limited access to modern farming technologies, and weak market connectivity. These constraints often result in low productivity, increased production costs, and unstable incomes. The lack of timely information and scientific decision-making tools further restricts farmers' ability to adopt efficient and sustainable agricultural practices.

In the floriculture sector, especially in jasmine cultivation, these challenges are intensified due to the crop's high sensitivity to diseases at early growth stages, bud infections, and flower damage. Maintaining quality and yield requires precise timing in disease detection and treatment. However, traditional methods of manual inspection are time-consuming, labor-intensive, and highly dependent on individual experience. Such methods are prone to human error and often lead to delayed diagnosis or inappropriate treatment, which can significantly reduce market value and overall profitability.

To overcome these limitations, this project proposes a specialized "Smart Monitoring of Jasmine Plant Growth, Disease Detection, and Yield Improvement" system. By integrating Artificial Intelligence (AI), Machine Learning (ML), computer vision, and Deep Learning(DL), the system offers a comprehensive and technology-driven solution tailored to jasmine cultivation. It enables crop-specific intelligence, multi-stage disease identification, automated severity assessment, and accurate yield prediction. Unlike generic agricultural platforms, this system focuses on the unique requirements of jasmine farming, empowering farmers with real-time insights and data-driven recommendations to improve productivity, reduce losses, and promote sustainable cultivation practices.

2. Problem Statement

Jasmine cultivation plays an important role in supporting the livelihoods of many farmers, especially in floriculture dependent regions. However, jasmine farmers frequently suffer significant losses in both yield and quality due to the late detection of plant diseases. Most disease identification is still carried out through manual visual inspection, which depends heavily on individual experience and availability of skilled labor. As a result, early symptoms are often overlooked, allowing infections to spread rapidly and cause extensive crop damage.

The traditional manual diagnosis process is inherently slow, subjective, and prone to human error. Variations in lighting conditions, plant growth stages, and observer expertise can lead to inconsistent assessments. In many cases, incorrect or delayed diagnosis results in the application of unsuitable pesticides or treatments, increasing production costs and negatively affecting plant health. Such practices not only reduce crop productivity but also pose risks to environmental sustainability and long-term soil fertility.

To overcome these limitations, there is a pressing need for an efficient Machine Learning–based disease detection system for jasmine plants. An automated system can analyze plant images in real time, identify disease patterns with high accuracy, and recommend appropriate control measures. By enabling rapid and reliable diagnosis, this technology can help farmers take timely preventive actions, minimize crop losses, improve overall production quality, and promote sustainable and profitable jasmine cultivation.

3. Proposed Solution

The proposed **"Smart Monitoring of Jasmine Plant Growth, Disease Detection, and Yield Improvement"** system is a comprehensive, multi-faceted platform designed to empower various stakeholders in the jasmine farming ecosystem, including farmers, field supervisors, and agricultural experts. It aims to provide precise, timely, and actionable insights to improve crop health, optimize resource utilization, and enhance overall yield. By integrating advanced technologies such as Artificial Intelligence (AI), Machine Learning (ML), computer vision, and Deep Learning (DL), the system enables continuous monitoring of plant conditions and environmental factors.

In addition, the platform supports early disease detection, automated severity assessment, and AI-based recommendations for treatment and crop management. It also assists farmers in making informed decisions related to irrigation, fertilization, and pest control. Through data-driven analysis and real-time monitoring, the system helps reduce losses, improve productivity, and promote sustainable and profitable jasmine cultivation.

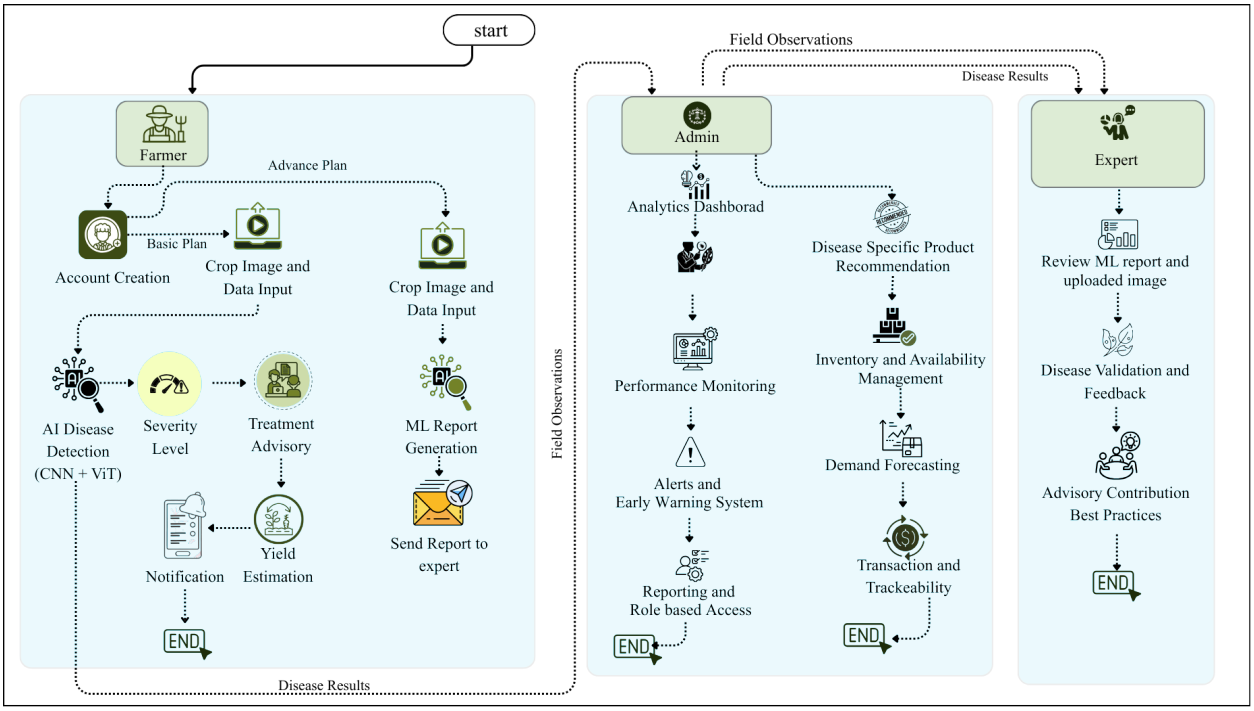
The core functionalities of the system include:

1. **User Registration & Secure Authentication:** This feature enables farmers, experts, and administrators to securely register and log in to the application using JWT-based authentication. It ensures password encryption, token validation, and role-based access control to protect user data and restrict unauthorized access.
2. **Farmer Profile & Farm Management:** Farmers can create and manage their profiles by adding details such as crop type, land size, season, and location. The system allows crop selection and farm setup to provide personalized advisories. Farmers can also log activities such as irrigation, spraying, and harvesting for better farm monitoring and record maintenance.
3. **AI-Based Image Disease Detection:** This feature allows farmers to upload crop images through the Android application for automated disease and pest detection. The system uses a machine learning/deep learning model to analyze images and predict the disease type, enabling early identification of crop health issues.
4. **Disease & Pest Advisory System:** After detecting a disease, the system provides detailed advisory information including disease name, symptoms, recommended treatments, preventive measures, and best practices. This helps farmers take immediate and appropriate action to reduce crop loss.
5. **Weather-Based Alerts & Predictive Risk Monitoring:** The system integrates real-time weather data (rainfall, temperature, humidity) to generate alerts and assess potential risks. It performs predictive analysis using historical and environmental data to provide early warnings about possible disease outbreaks or weather-related threats.

6. **Admin Dashboard & Monitoring:** The admin module allows administrators to monitor registered farmers, view disease distribution and severity analysis, generate summary reports, and manage overall system data. Admins can review farmer submissions and forward compiled reports to experts for validation.
7. **Expert Review & Verification Module:** Experts or specialists can access submitted reports, review AI-generated recommendations, edit advisories if necessary, and verify final reports. The system maintains report status tracking (Draft, Sent, Reviewed, Verified) and records expert validation details.
8. **Notification System:** The platform sends push notifications to farmers for disease alerts, weather warnings, expert verification updates, and important advisories using a cloud messaging service.
9. **Government Scheme Information:** The system provides information about relevant agricultural schemes and subsidies based on crop type and region, helping farmers stay informed about available benefits and support programs.
10. **Offline Support & Multilingual Access:** The application offers limited offline functionality by caching essential advisories and synchronizing data when the internet connection is restored. It also supports multiple regional languages to enhance accessibility and usability.
11. **Basic Analytics & Reports:** The system generates summaries related to crop health, advisory trends, and farm activity logs. Both mobile and web platforms are synchronized through a centralized database to ensure real-time data consistency across devices..

4. Methodology / Block Diagram

The system's architecture is designed to support a comprehensive workflow involving multiple stakeholders: Farmers, Admin (Agri-Tech Company Personnel), and Experts (Agronomy Consultants). The core flow revolves around AI-driven crop disease detection and integrates multiple modules to ensure accurate diagnosis, advisory support, monitoring, and operational management.



Overall System Flow

- **Start:**The process initiates with Account Creation by the Farmer through either a Basic Plan or Advance Plan subscription.

1. **Farmer Input (Crop Image & Field Data)**
    a. **Farmers:**
        i. Capture and upload images of hibiscus plants (all supported types from Karnataka as defined in the system).
        ii. Provide relevant crop details such as growth stage, visible symptoms, and optional environmental data.
        iii. Submit the data for AI-based analysis.
2. **AI Disease Detection (CNN + Vision Transformer)**
    a. **Image Preprocessing:**
        i. Noise removal and resizing
        ii. Leaf segmentation (if required)

iii. Background isolation

b. Feature Extraction:

i. Advanced deep learning models extract high-level visual features:
ii. CNN architectures
iii. Vision Transformer (ViT)
iv. Hybrid CNN + ViT model

c. Disease Prediction:

i. Classification identifies the specific disease.
ii. Severity Level estimation (Low / Medium / High).

d. Advisory Generation:

i. Disease-specific treatment recommendations.
ii. Pesticide or organic solution suggestions.
iii. Preventive measures for early-stage infection.

e. Additional Outputs:

i. Yield Estimation based on severity.
ii. ML Report Generation summarizing:
    1. Disease type
    2. Severity
    3. Recommended treatment
iii. Notification sent to farmer with results.

3. Subscription-Based Expert Review

a. Advance Plan Workflow:

i. If the farmer subscribes to the Advance Plan:
    1. The ML report and uploaded images are forwarded to the Expert (Agronomy Consultant).

b. Expert Workflow:

i. Log into the Expert Panel.
ii. Review ML reports and crop images.
iii. Validate or correct disease prediction.
iv. Provide enhanced advisory and best practices.
v. Submit expert feedback to improve system intelligence.
vi. The validated report is then shared with the farmer.

c. **Basic Plan Workflow**:

i. Farmer receives only the AI-generated ML report.
ii. No expert validation is included.

4. Admin Module (Agri-Tech Company Operations)

The Admin oversees system-level intelligence and management.

a. **Analytics Dashboard:**
    i. Monitor disease trends.
    ii. Track farmer usage statistics.
    iii. Analyze regional outbreak patterns.

b. **Performance Monitoring:**
    i. Evaluate model accuracy.
    ii. Track system performance metrics.

c. **Alerts & Early Warning System:**
    i. Detecting disease clusters.
    ii. Send outbreak alerts to affected regions.

d. **Disease-Specific Product Recommendation:**
    i. Map disease type → recommended products.

e. **Inventory & Availability Management:**
    i. Track product stock levels.
    ii. Monitor supplier availability.

f. **Demand Forecasting:**
    i. Predict pesticide and treatment demand.
    ii. Estimate regional supply requirements.

g. **Transaction & Traceability:**
    i. Track product distribution.
    ii. Maintain transparent supply records.

h. **Reporting & Role-Based Access:**
    i. Manage Farmer, Expert, and Admin roles.
    ii. Generate operational and analytical reports.

5. Hardware, Software and Tools Requirements

The proposed "Smart Monitoring of Jasmine Plant Growth, Disease Detection, and Yield Improvement" system is developed as a fully software-based, cloud-enabled intelligent platform. It focuses on image-based disease detection, data analytics, and decision support without direct hardware integration. The system uses modern web technologies, machine learning frameworks, and cloud infrastructure to ensure scalability, reliability, and high performance.

A. Data Collection Layer

This layer handles input data provided by users and external sources.

1. **Image Upload:**
   Farmers upload images of jasmine leaves, buds, and flowers through mobile or web applications.
2. **Manual Input:**
   Users enter crop-related data such as location, soil type, growth stage, and cultivation practices.
3. **External APIs:**
   Weather, market price, and water resource data are collected using third-party APIs.

This approach enables flexible data acquisition without the need for physical sensors.

B. Communication Layer

The communication layer ensures secure and efficient data exchange between system components.

1. **Protocols Used:**
   a. HTTP / HTTPS
   b. RESTful APIs

These protocols enable smooth interaction between frontend applications, backend servers, and machine learning services.

C. Backend Layer

The backend layer manages business logic, data processing, and service integration.

- **Frameworks Used:**
  - FastAPI / Flask (Python)

Functions include:

1. Receiving user inputs and images
2. Forwarding data to ML models
3. Managing predictions and reports
4. Handling authentication and authorization
5. Providing APIs for frontend access

FastAPI is preferred for high-performance applications, while Flask offers simplicity for rapid development.

D. Machine Learning and Image Processing Layer

This layer performs intelligent analysis and prediction.

Frameworks and Libraries

1. TensorFlow / PyTorch
2. OpenCV
3. Scikit-learn

Model Components

1. Image Preprocessing

1. Noise removal
2. Resizing
3. Contrast enhancement
4. Segmentation

2. Feature Extraction

1. Inception V3
2. VGG19
3. MobileNetV3
4. Vision Transformers (ViT)
5. Convolutional Neural Networks (CNN)

3. Disease Classification

1. CNN with Softmax
2. Support Vector Machine (SVM)
3. Random Forest

4. Yield and Price Prediction

1. ARIMA
2. LSTM

These models provide accurate disease identification, severity estimation, and yield forecasting.

E. Data Processing and Rules Engine

This layer converts raw data into actionable insights.

- **Tools Used:**
  Python with Pandas and NumPy

Functions include:

1. Environmental data analysis
2. Risk assessment
3. Advisory generation
4. Fertilizer and irrigation recommendations
5. Crop management rules

The rules engine supports automated decision-making.

F. Database Layer

This layer stores structured and unstructured data.

1. **Databases Used:**
   MongoDB / Firebase Firestore

Data stored includes:

1. User profiles
2. Uploaded images metadata
3. Disease records
4. Prediction history
5. Market and weather data

These databases provide scalability and high availability.

G. Frontend Layer

This layer provides user interaction interfaces.

Mobile Application

- Flutter (Dart)
- Provider / Riverpod

Features:

1. Image upload
2. Disease reports
3. Alerts
4. Advisory dashboard

Web Application

1. React.js
2. Tailwind CSS
3. Redux Toolkit / Context API

Features:

- Admin dashboard
- Data visualization
- User management
- System monitoring

I. Authentication and Security Layer

This layer ensures secure system access.

1. **Technologies Used:**
   JWT
   Firebase Authentication

Security measures include:

1. Encrypted communication (HTTPS)
2. Role-based access control
3. Secure token validation

J. Notification System

This layer provides real-time alerts.

1. **Service Used:**
   Firebase Cloud Messaging (FCM)

Notifications include:

1. Disease warnings
2. Treatment reminders
3. Weather alerts
4. Market updates

K. External API Integration Layer

This layer connects external information services.

Weather Data

1. OpenWeatherMap API
2. IMD API

Market Prices

1. e-NAM API

Mapping and GIS

1. Google Maps API / Mapbox
2. GeoJSON

These services support location-based and market-driven decision-making.

L. Deployment and Hosting Layer

This layer manages system deployment and scalability.

Cloud Platforms

1. AWS / Google Cloud Platform / Microsoft Azure

Model Deployment

1. TensorFlow Serving / TorchServe / FastAPI ML APIs

Used for:

1. Automated testing
2. Continuous deployment

6. Proposed Evaluation Measures

To ensure the effectiveness, accuracy, and reliability of the proposed **"Smart Monitoring of Jasmine Plant Growth, Disease Detection, and Yield Improvement"** system, multiple quantitative and qualitative evaluation parameters will be adopted. These measures will assess the performance of the machine learning models, system responsiveness, user satisfaction, and overall impact on agricultural productivity.

1. **Accuracy** will be used to evaluate the overall correctness of the system in disease detection and yield prediction. It represents the proportion of correct predictions made by the model and reflects the general performance of the system.
2. **Precision** will measure the proportion of correctly identified diseased plants among all predicted diseased cases. This helps in minimizing false positives and avoiding unnecessary treatments and expenses.
3. **Recall** will assess the system's ability to correctly identify actual diseased plants. High recall is important for early disease detection and reduces the chances of missing infected crops.
4. **F1-Score** will be used as a balanced performance measure by combining precision and recall. It is especially useful when dealing with imbalanced datasets.
5. **Confusion Matrix** will be employed to visualize classification performance by showing true positives, true negatives, false positives, and false negatives.
6. **Intersection over Union (IoU)** will be applied to evaluate segmentation accuracy by measuring the overlap between predicted and actual diseased regions.
7. **Inference Time** will be measured to evaluate the speed of prediction, which is essential for real-time field applications.
8. **ROC-AUC** will be used to analyze the model's ability to distinguish between healthy and diseased plants under different threshold settings.
9. **User Acceptance and Satisfaction** will be assessed through feedback from farmers and experts, focusing on usability, clarity, and trust in recommendations.
10. **Economic Impact Analysis** will evaluate reductions in crop loss, improvements in yield, and increases in market value.
11. **Scalability and Robustness** will be tested under different conditions and user loads to ensure stable and reliable system performance.

7. Conclusion

The **“Smart Monitoring of Jasmine Plant Growth, Disease Detection, and Yield Improvement”** system addresses major limitations in traditional and existing digital agricultural practices, especially in specialized floriculture such as jasmine cultivation. Conventional methods often lack timely disease detection, accurate diagnosis, and integrated decision-support mechanisms. By incorporating advanced Artificial Intelligence (AI) and Machine Learning (ML) models, Deep Learning (DL) models, and multi-platform interfaces, the proposed system provides a comprehensive solution for early disease identification, precise treatment recommendations, and informed decision-making related to yield improvement and market access.

The proposed system architecture is designed with dedicated workflows for farmers, field supervisors, and agronomy experts to ensure efficient communication and effective utilization of generated insights. This structured approach enables seamless sharing of real-time data and expert recommendations. The system includes features such as disease severity estimation, yield prediction, fertilizer optimization, smart irrigation management, and climate risk alerts. These functionalities help farmers move from traditional reactive practices to proactive and preventive crop management strategies.

Overall, the integrated and technology-driven nature of the system is expected to significantly reduce crop losses, improve jasmine yield and quality, and minimize operational costs. By promoting timely interventions and optimal resource utilization, the system enhances farm productivity and profitability. Furthermore, its focus on crop-specific requirements and data-driven decision-making contributes to sustainable agricultural practices and supports the long-term economic well-being of jasmine growers in India.